

Part 1 of 2

Power over Ethernet For Networked Devices

Power over Ethernet (PoE) is an emerging new technology that delivers power and data on an Ethernet cable at the same time. Here we discuss this technology and concerns in its deployment, especially in industrial environments

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Power over Ethernet (PoE) is modeled after the technology used in telecom systems that supplies power to telephones. Thus it powers network devices without the need of AC electrical outlets. By eliminating the power cords and the power infrastructure, it greatly simplifies the deployment and brings lots of convenience and benefits to homes, offices and industrial environments.

The IEEE802.3af standard describes the mechanism for power over Ethernet. The technology relies on a form of access called 'carrier sense multiple access with collision detection' (CSMA/CD) and physical layer specification. Released in 2003, the standard was entitled 'data terminal equipment (DTE) power via media dependent interface.' It defines how to integrate power into a data cable while ensuring data integrity and safety of the network.

As shown in Fig. 1, a typical PoE installation involves two major components, the power sourcing equipment (PSE) and the powered device (PD). All the powered devices are PoE-compatible and accept power transmitted over a standard CAT5 cabling.

Normally, the PSE supplies 48V with a maximum current of 350 mA. This power passes from the PSE to PDs via CAT5 cable. With some loss in the cable, the supply is limited to 13W to drive various kinds of devices,



Fig. 1: Typical PoE installation

such as VoIP phones, wireless access points, IP security cameras and bar code readers.

How is power delivered through the cable?

PoE transmits data and power on Ethernet cable at the same time.

There are four unshielded twisted pairs in a CAT5 Ethernet cable. Only two of the pairs are used in 10BASE-T and 100BASE-T environments. To integrate power and data over the cable, IEEE802.3af defines two alternatives for different circumstances:

1. Alternative A. It is also known as phantom powering. In it, a PSE feeds power over data pin pairs 1-2 and 3-6 without disturbing data transfer.

2. Alternative B. The power is delivered through the idle pairs; pins 4 and 5 used are for positive

supply and pins 7 and 8 for negative supply.

According to the standard, the PSE can be designed to apply power either by alternative A or alternative B, but not both at the same time. As a result, a powered device must be capable of receiving power from both options while the source may apply power in either way.

Power sourcing equipment

As is clear from above, a PSE supplies power to the powered device. PoE-capable switches are the most common examples of PSEs.

Acting as a power transmitter, the PSE has three main jobs:

1. Detect a PD and determine its power level
2. Supply adequate power according to the power level
3. Monitor and stop power supply

Two types of PSEs are defined in the standard, the endspan and the midspan. An endspan PSE is a PoE-capable port that carries both data and power on the link, while a midspan PSE stands between a common Ethernet port and a PD to help inject power. Midspan offers a solution when adding PoE to an existing network infrastructure without changing its original configuration.

Powered devices and their classes

Powered devices receive power from the PSE. Today, more and more